In all my science interpretations of what happens in *Interstellar*, I assume that this actually *is* Gargantua's mass: 100 million Suns.[15] For example, I assume this mass in Chapter 17, when explaining how Gargantua's tidal forces could produce the giant water waves that inundate the Ranger on Miller's planet.

The circumference of a black hole's event horizon is proportional to the hole's mass. For Gargantua's 100 million solar masses, the horizon circumference works out to be approximately the same as the Earth's orbit around the Sun: about *1 billion kilometers*. That's big! After consulting with me, that's the circumference assumed by Paul Franklin's visual-effects team, when producing the images in *Interstellar*.

Physicists attribute to a black hole a radius equal to its horizon's circumference divided by 2π (about 6.28). Because of the extreme warping of space inside the black hole, this is not the hole's true radius. Not the true distance from the horizon to the hole's center, as measured in our universe. But it *is* the event horizon's radius (half its diameter) as measured in the bulk; see Figure 6.3 below. Gargantua's radius, in this sense, is about 150 million kilometers, the same as the radius of the Earth's orbit around the Sun.

Gargantua's Spin



When Christopher Nolan told me how much slowing of time he wanted on Miller's planet, *one hour there is seven years back on Earth*, I was shocked. I didn't think that possible and I told Chris so. "It's non-negotiable," Chris insisted. So, not for the first time and also not the last, I went home, thought about it, did some calculations with Einstein's relativistic equations, and found a way.